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Object Oriented Programming
BSIT 2-3

Midterm Assignment #1 - Introduction to Java

1. What is Java Programming language? Discuss its key features and advantages.

Developed by Sun Microsystems in 1995 and later acquired by Oracle Corporation, Java is a high-level, object-oriented programming language designed to be reliable programming language for coding everything from software application to enterprise level application.

Java is based on objects and classes. It, supports encapsulation, inheritence, and polymorphism. It is also platform independent meaning it can be run anywhere in any operating system. It is secure, robust, distributed, and high performance since it uses JIT or just in time compiler.

2. Explain the concept of platform independence in Java. How does Java achieve platform independence?

Platform independence in Java means it can run on any operating systems without needing to be modified or recompiled. Java source code is compiled into an intermediate form called bytecode. Bytecode is not executed directly by the operating system but is run by Java Virtual Machine (JVM). Since every operating systems have their own JVM, the same bytecode can run on any system that has a compatible JVM. Instead of compiling into machine-specific code, java compiles into platform-neutral bytecode.

3. Discuss the main components of a Java program and their roles.

A java program has the following components: Class which acts as a blueprint of objects. Every Java program must have atleast one class. It contains variables to store data along with methods to define behavior. There is also variables which is

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used to store data values. There is also methods which is divided into instance method that work with the object-specific data and static methods that belong to the class as a whole. Basically, it is functions defined inside a class to perform tasks. There is also statement that carries out instructions, packages which help organize related classes and interfaces. Objects, that represent entities within the program. Every program has public static void main (String[] args) method, which is were the execution starts. It is called, Main Method. Each component of Java works together to create a functional Java application.

4. what are the diffences between JDK, JRE, and JVM in Java?

Java Development Kit (JDK) is a software development package used to create Java applications. It includes compilers, debugger, and utilities along with the JRE. Java Runtime Environment provides necessary libraries and JVM to run Java programs, however it does not include tools for development. Java Virtual machine (JVM) is an execution engine that converts Java bytecode into machine-specific instructions, handles memory management, and garbage collections.

5. Explain the difference between interpreted and compiled languages. Is Java an Interpreted or Compiled language?

Compiled language is one that which source code is translated by the compiler into machine code before execution, allowing program to run directly on CPU resulting in faster performance. Interpreted language on the other hand, executes language or instructions directly through an interpreter without producing a separate machine-code file (executable). This makes debugging easier but makes it slower in execution.

In the case of Java, it is both compiled and interpreted language. It is because its source code is compiled into bytecode by compiler and then that bytecode is interpreted by the JVM during execution